

ZS-A15

Pre-Registered Gaia DR4 Falsification Protocol for the ε -Halo Outer Boundary: Enclosed-Mass Saturation, Keplerian Slope, and the NFW/MOND Degeneracy

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Dependencies: ZS-F1, ZS-A1, ZS-A5, ZS-A11 v1.1.

Frozen protocol: a15_dr4_protocol.yaml | SHA-256

1d04ee2f73b6f7791586fa3964d201e2336e54f690a75fc25f3ee0f773b283e2

GitHub: <https://github.com/KennyKang-git/zspin>

Verification Summary

Verification: 36/36 PASS | Zero free parameters in the structural claim (a finite outer cutoff exists). The value r_Z is DERIVED-CONDITIONAL on the observed inputs (M_{total} , v_{flat}) and is registered, not predicted ab initio. Sole geometric inputs: $A = 35/437$, $Q = 11$, $(Z, X, Y) = (2, 3, 6)$. Pre-registered before Gaia DR4 (2 December 2026); the frozen analysis protocol is hash-locked (SHA-256 above). The verification suite includes a five-truth confusion-matrix (T20a–e) and a zeroth-order transition-width measurement (T23a–c).

§0. Abstract

The ε -Halo of Z-Spin Cosmology models galactic dark matter as the gradient energy of the exactly massless Goldstone mode θ of a spontaneously broken internal $U(1)_Z$, giving $\rho_\theta \propto 1/r^2$ — functionally identical to a singular isothermal sphere and, across seven macroscopic observables, to a Λ CDM sub-halo (ZS-A11 v1.1). The descriptions break only at microscopic ontology (the one-way detection asymmetry F-A5.7) and at the outer boundary, where the Three-Region structure of ZS-F1 §5.3 terminates the halo at a finite radius r_Z (DERIVED).

This is not a claim that Z-Spin has measured the Galaxy; it is a pre-registered, hash-locked falsification protocol fixed before Gaia DR4. We freeze a single decisive Milky Way test: whether the enclosed mass saturates near $M_{\text{total}} \approx 2 \times 10^{11} M_\odot$ by ≈ 25 kpc (logarithmic slope $\gamma \rightarrow -1/2$) versus continuing to grow toward $M_{200} \approx 8 \times 10^{11} M_\odot$. Four honesties are built in. (i) A finite cutoff is DERIVED upstream and shared by every finite-mass halo (truncated or cored NFW included); it is not advanced as novel (§2A). (ii) The value r_Z is conditional on observed inputs (DERIVED-CONDITIONAL); the derivation gap is split into the now-closed well-posedness of the boundary-value problem (A15-T1a, IMPORTED-PROVEN via Ginzburg–Landau / harmonic-map theory) and the transition width. We prove a No-Intrinsic-Width Lemma: the strictly massless equation $\square\theta = 0$ fixes the logarithmic profile but determines no universal width coefficient, so the observed $\Delta r/r_Z \lesssim 0.4$ (an upper bound, A15-T1b-0) cannot come from the bulk equation; the v1.2 geometric candidates $\{A, 1/Q, Z/Q\}$ are accordingly RETRACTED, and a smooth massless-edge ansatz is shown to be both non-harmonic and $\approx 5\times$ too

gradual. The open question is reframed as a Sharp-Edge Mechanism (A15-T1b-3): a scale census shows no locked length equals ≈ 0.4 , so the edge must be short-scale (sharp). The corpus matching $M_{\varepsilon}(r_Z) = M_{\text{total}}$ is a global point boundary condition, which on the scale-free interior yields a near-step edge (DERIVED-CONDITIONAL, no new parameter) — turning the v1.4 “data sharper than the smooth field” tension into a prediction, and registering a new falsifier (F-A15.7: a wide DR4-resolved intrinsic width would refute it). No numerical $\Delta r/r_Z$ is claimed. (iii) The protocol does not exclude all particle dark matter and, on the rotation curve alone, does not separate ε -Halo from the Moffat-type MOG family; this is structured as a four-model decision matrix (§4B) and a five-truth confusion-matrix verification (T20a–e), not hidden. (iv) An Imported No-Go Ledger (§2A) records, with external citations, exactly where the rotation curve alone cannot decide. The present record is contested (Jiao et al. 2023 low-mass versus Zhou et al. 2022 full-halo), which is why advance registration converts a possible post-DR4 agreement from numerology into evidence.

Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem; standard mathematics alone, machine-verifiable.
DERIVED	Z-Spin action + standard physics, zero free parameters.
DERIVED-CONDITIONAL	DERIVED conditional on a listed observational input or upstream theorem.
IMPORTED-PROVEN	Result proved externally and used here without re-proof; full citation given.
IMPORTED-NO-GO	External result that bounds what the present data can decide.
TESTABLE	Pre-registered prediction with an explicit falsification protocol.
HYPOTHESIS-weak	Motivated conjecture with a partial derivation chain; promotion path noted.
OBSERVATION	Empirical regularity; anti-numerology controlled; origin/discrimination pending.
LOCKED	Core constant fixed upstream; no downstream paper may modify it.
NON-CLAIM	Explicit declaration of what is NOT asserted; bounds the framework’s scope.
OPEN	Recognized gap honestly registered for future work.
OBSERVATION-upper-bound	Empirical quantity that bounds, but does not measure, the underlying value.
HYPOTHESIS-strong	Conjecture with a clear derivation target and a plausible mechanism identified.
RETRACTED-in-session	Previously registered claim withdrawn within this work after new evidence.
DERIVED-interpretation	Qualitative reading of a derived structure; not a quantitative prediction.

§1. Introduction

In Z-Spin Cosmology the flat rotation of disk galaxies is produced not by massive particles but by the gradient energy of a Goldstone field. The Z-anchor theorem (ZS-F1 §5.2, PROVEN) forces $|\Phi| = 0$ at any point of non-zero winding from $\pi_1(U(1)) = \mathbb{Z}$; around the resulting vortex line the angular mode θ is the exactly massless Goldstone boson (ZS-F1 §4.2, DERIVED) with profile $\theta(r) = \ln(r/r_0)/L$, giving the isothermal energy density of equation (3). ZS-A11 v1.1 showed this ε -Halo is degenerate with a Λ CDM sub-halo across profile, source, and seven observables, breaking only at microscopic ontology and at the outer boundary.

Gaia DR4 is scheduled for 2 December 2026 (first 5.5 years / 66 months of mission astrometry, epoch data, improved proper motions). Because a prediction fixed after the data carries no evidential weight, this paper freezes the discriminating prediction, its pipeline, and its decision rules in advance, binding them with a cryptographic hash (§4A). Section 2A imports the external theorems and no-go results that bound the test; §3 separates what is already derived from the one quantity DR4 genuinely puts at risk.

§2. Locked Inputs and External Proven Lemmas

All Z-Spin inputs are LOCKED upstream; no new parameter is introduced. External lemmas are standard galactic dynamics, quoted as PROVEN.

Table 2.1. Locked inputs and their sources.

Symbol	Value	Source / Status
A	35/437 = 0.080092...	ZS-F2 geometric impedance — LOCKED
Q	11 (prime)	ZS-F5 register dimension — PROVEN
(Z, X, Y)	(2, 3, 6)	ZS-F5 sector decomposition — PROVEN
M_P, ℓ_P	1.221×10 ¹⁹ GeV, 1.616×10 ⁻³⁵ m	CODATA 2022 — STANDARD
M_total (MW)	2.06 (+0.24, -0.13)×10 ¹¹ M☉	Jiao et al. 2023 — OBSERVATION
v_flat (MW)	220 – 237 km/s	Jiao 2023; Cepheid 2025 — OBSERVATION

Lemma 1 (SIS profile and velocity, PROVEN).

For a self-gravitating isothermal system [5], $\rho_{\text{SIS}} = \sigma^2 / (2\pi G r^2)$, $M(<r) = 2\sigma^2 r / G$, $v_c^2 = G M(<r) / r = 2\sigma^2$; hence $v_{\text{flat}} = \sqrt{2} \cdot \sigma$. The MW’s observed 220 km/s is the circular velocity, so the equivalent SIS dispersion is $\sigma \approx 156$ km/s (this closes the $\sigma \leftrightarrow v_{\text{flat}}$ normalization of ZS-A11 §3.2; it affects L only, not r_Z).

Lemma 2 (Finite-region enclosed mass, PROVEN).

In the flat region $M(<r) = v_{\text{flat}}^2 r / G$ grows linearly and terminates at $r_Z = G M_{\text{total}} / v_{\text{flat}}^2$ (2), elementary Newtonian dynamics and framework-independent.

Lemma 3 (Cutoff necessity, PROVEN).

A profile $\rho \propto r^{-\alpha}$ with $\alpha \leq 3$ has divergent total mass [5]; the $\alpha = 2$ law must truncate. A finite cutoff is a theorem of classical dynamics, shared by the isothermal halo, truncated NFW, and any finite-mass model alike.

§2A. Imported Theorem and No-Go Ledger

To prevent cherry-picking, we fix — with external citations — both the theorems we rely on and the boundaries beyond which the rotation curve alone cannot decide. The no-go rows are not weaknesses to be hidden; they are the scope of the test.

Table 2A.1. Imported theorems (I) and no-go results (N).

ID	Imported result	A15 consequence	Status
I1	Finite-mass isothermal no-go: $\rho \propto 1/r^2$ cannot persist to infinity [5]	the halo must have a finite cutoff r_Z	IMPORTED-PROVEN

ID	Imported result	A15 consequence	Status
I2	Exterior point-mass theorem: M saturates $\Rightarrow v_c \propto r^{-1/2}$ [5]	saturation $\Rightarrow$ Keplerian $\gamma = -1/2$	PROVEN
I3	Annular winding BVP well-posed (Ginzburg–Landau / S^1 harmonic map, prescribed degree) [13–15]	the θ -transition BVP (A15-T1a) is a normal, solvable problem	IMPORTED-PROVEN
N1	Ou et al. cored-Einasto fit to the MW decline [7]	a decline alone does NOT falsify particle DM	IMPORTED-NO-GO
N2	MOG fit to the declining MW curve [9]	ε -Halo vs MOG unresolved by the rotation curve alone	IMPORTED-NO-GO
N3	Disk–halo (maximum-disk) degeneracy [16]	an independent mass channel is required	IMPORTED-NO-GO
N4	Klacka–Sturc axisymmetric-Jeans critique [10]	spherical/axisymmetric consistency clause is mandatory	IMPORTED-NO-GO
N5	Transition radius is model-dependent: MOG $R_0 = 18.1$ kpc vs Newtonian $R_0 = 33.8$ kpc on the same four DR3 datasets [9]	Δr must be read from a model-independent slope break, not a fitted R_0	IMPORTED-NO-GO
N6	FIRE-2 synthetic-Gaia Jeans systematics 5–20%, large galaxy-to-galaxy variance [17]	the zeroth-order Δr is a loose band; precise quantization needs DR4	IMPORTED-NO-GO

§3. What Is Already Derived versus What Is Registered

The Z-Spin halo energy density is $\rho_\theta(r) = M_P^2/(2 L^2 r^2)$ (3), the gradient energy of the Goldstone solution (ZS-F1 §4.3, DERIVED); equating with Lemma 1 gives $L^2 = \pi G M_P^2/\sigma^2$, L of order the Hubble radius.

Table 3.1. Status ledger: what this paper does and does not claim.

Statement	Status	Source
$\rho_\theta \propto 1/r^2$ from the massless Goldstone mode	DERIVED	ZS-F1 §4.3
Z-anchor $ \Phi = 0$ from $\pi_1(U(1)) = \mathbb{Z}$	PROVEN	ZS-F1 §5.2
A finite outer cutoff r_Z exists	DERIVED	ZS-F1 §5.3
Existence of a cutoff is novel to Z-Spin	NON-CLAIM	I1, Lemma 3
Equivalence with sub-halo across 7 observables	DERIVED	ZS-A11 §3
One-way detection asymmetry F-A5.7	PROVEN	ZS-A5
Discriminating DR4 protocol (saturation vs growth)	TESTABLE	§4, §4A
Numerical value $r_Z \approx 18$ kpc	DERIVED-COND.	Eq. (2)+obs.
θ -transition BVP is well-posed (A15-T1a)	DERIVED-COND.	§5, I3
$\square\theta=0$ fixes no width coefficient (No-Intrinsic-Width)	PROVEN	§5, T24
Transition width $\Delta r/r_Z \lesssim 0.4$ (A15-T1b-0)	OBS-upper-bnd	§5, T23a
Geometric Δr candidates $\{A, 1/Q, Z/Q\}$	RETRACTED	§5, T23b
Intrinsic edge near-step (Path A, point matching)	DERIVED-COND.	§5, T25
Numerical $\Delta r/r_Z$ value	OPEN / unclaimed	§5
ε -Halo separated from MOG by RC alone	OPEN	§4B, N2

§4. Pre-Registered Prediction and Falsification Gates

The contested record. All recent analyses agree the outer curve declines but disagree on the implied mass. Jiao et al. (2023): sharp Keplerian onset from ≈ 19 kpc, $M_{\text{total}} = 2.06 \times 10^{11} M_{\odot}$. Zhou et al. (2022): milder decline, full halo $M_{200} = 8.05 \times 10^{11} M_{\odot}$ to $R_{200} \approx 192$ kpc. Ou et al. (2024): faster outer decline but a cored DM profile. Because $r_Z \propto M_{\text{total}}$, the ε -Halo reading holds only at low mass; at $M_{200} \approx 8 \times 10^{11} M_{\odot}$ equation (2) gives $r_Z \approx 71$ kpc. The decisive observable is the enclosed-mass behaviour at large radii.

Frozen statistic. $\gamma \equiv d \ln v_c / d \ln r$ (0 flat, $-1/2$ Keplerian). The gates below, and the full pipeline of §4A, are frozen and hash-locked before DR4.

Table 4.1. Pre-registered Gaia DR4 falsification gates.

Gate	Pre-registered condition and falsifier	Status
G-DR4.1	Structural. Enclosed mass saturates near $2 \times 10^{11} M_{\odot}$ by ≈ 25 kpc ($\gamma \rightarrow -0.5 \pm 0.1$, maintained). FALSIFY if $\gamma > -0.3$ beyond 25 kpc or $M(<r)$ grows toward $8 \times 10^{11} M_{\odot}$.	TESTABLE
G-DR4.2	Numerical. Onset r_Z in $[15, 21]$ kpc (propagated from $M_{\text{total}} = 1.93 - 2.30 \times 10^{11}$, $v_{\text{flat}} = 220 - 237$). FALSIFY if outside $[13, 25]$ kpc at 2σ .	DERIVED-COND.
G-DR4.3	Systematic. γ from spherical AND axisymmetric Jeans (N4). If the spherical treatment removes the decline, G-DR4.1/2 are N/A — not PASS.	TESTABLE

§4A. Frozen DR4 Analysis Protocol

The pipeline is fixed in `a15_dr4_protocol.yaml` (SHA-256 `1d04ee2f73b6f7791586fa3964d201e2336e54f690a75fc25f3ee0f773b283e2`); any post-release change alters the hash and voids the registration. Tracers: RGB/LRGB, classical Cepheids, RR Lyrae, BHB, K giants — none may be dropped to recover a slope. Radius bins (kpc): $[15, 20], [20, 25], [25, 30], [30, 40], [40, 60]$; γ is evaluated on the three outer bins (≥ 25 kpc). Estimators: at least two of axisymmetric Jeans, spherical Jeans, action-based DF, with the spherical-Jeans result mandatory (it directly tests N4). Decision statistics: γ_{outer} , dM/dr , M_{enclosed} ; gates G-DR4.1–G-DR4.3 are applied verbatim. The reference implementation `verify_zsa15.py` executes this rule: mock gate recovery (T17–T19) returns PASS / FAIL / N/A correctly, a hash check (T21) and a paper/code/protocol consistency check (T22) guard against silent edits.

§4B. Four-Model Decision Matrix and Confusion-Matrix Validation

The ε -Halo is not tested against particle-NFW alone. Within modified gravity the matrix distinguishes deep-MOND (asymptotically flat, hence disfavoured by a genuine decline) from the Moffat-type MOG family (running effective G , which fits a decline and stays degenerate with ε -Halo on the rotation curve).

Table 4B.1. Frozen DR4 outcome $\rightarrow$ model decision matrix.

DR4 outcome	ε -Halo	Particle NFW	Truncated / cored NFW	Modified gravity
$\gamma \approx -1/2$ maintained; M saturates $\approx 2 \times 10^{11}$	strongly supported	weakened	possible (tuned r_t)	MOG possible; MOND disfavoured
$\gamma \approx 0$ or M grows $\rightarrow 8 \times 10^{11}$	FALSIFIED	supported	supported	model-dependent

DR4 outcome	ε -Halo	Particle NFW	Truncated / cored NFW	Modified gravity
decline vanishes under spherical Jeans	N/A	N/A	N/A	N/A
decline + no lensing/dyn-mass mismatch	supported	weakened	possible	possible
decline + MOG fits equally	not sole winner	weakened	possible	MOG supported

Confusion-matrix validation. The verification suite implements this matrix as a five-truth Bayesian-information-criterion comparison (T20a–e): a saturated truth selects the sharp-decline class (ε -Halo wins over NFW and flat); an $M_{200} = 8 \times 10^{11}$ NFW truth selects NFW (ε -Halo loses); a cored mild-decline truth selects the cored model (ε -Halo loses); a MOG-decline truth leaves ε -Halo and MOG within $\Delta\text{BIC} < 2$ (UNRESOLVED, $|\Delta\text{BIC}| = 1.34$); and a spherical-Jeans artifact returns N/A. The classifier therefore changes verdict with the truth model — it is not tuned to let ε -Halo win.

§5. Open Gate A15-T1: Finite θ -BVP Closure

The weakest link in the value prediction is that equation (2) is a Newtonian identity in which M_{total} is fitted from the curve, so $r_Z \approx \text{onset}$ is near-tautological. We split the route to removing this conditionality into two sub-gates.

A15-T1a — BVP mathematical admissibility (IMPORTED-PROVEN / DERIVED-CONDITIONAL).

The transition profile is the solution of a winding-phase boundary-value problem on an annular region, $\theta \sim \ln r$ ($r \ll r_Z$), $\partial_r \theta \rightarrow 0$ ($r \rightarrow r_Z$), $\theta \rightarrow \text{const}$ ($r > r_Z$) (5), with FRW matching as the outer condition. This falls within the standard class of prescribed-degree Ginzburg–Landau / S^1 -valued harmonic-map problems, for which existence of a rotationally invariant minimizer and convergence to the harmonic-map solution are established theorems [13–15]. The BVP is therefore a normal, well-posed problem; A15-T1a is closed (IMPORTED-PROVEN), so the residual uncertainty is purely the Z-Spin-specific numerical scale, not the mathematical legitimacy of the construction.

No-Intrinsic-Width Lemma (PROVEN).

On a source-free annular domain the strictly massless equation $\square\theta = 0$ has the unique rotationally symmetric solution $\theta(r) = a \cdot \ln r + b$ (the radial Laplace ODE $(1/r)(r\theta')' = 0$ integrates to $\theta' = a/r$). This solution is scale-free: it fixes the logarithmic interior profile and the $1/r^2$ halo, and it admits a finite outer boundary-matching problem (A15-T1a), but it contains no length scale that could set a universal transition-width coefficient. Therefore any value such as $\Delta r/r_Z \approx 0.4$ must originate in the matching operator, finite vortex-line geometry, boundary regularization, or the observational inference pipeline — not in the bulk massless equation itself. This is why A15-T1b is OPEN, and it is a theorem, not a gap in effort (gate T24).

A15-T1b-0 — observational width proxy (OBSERVATION-upper-bound).

Existing Gaia DR3 analyses provide a coarse proxy, not a sharp intrinsic width. Jiao et al. (2023) place the roll-off from flat to Keplerian between ≈ 19.5 and ≈ 26.5 kpc, with the enclosed mass essentially constant ($1.9\text{--}2.0 \times 10^{11} M_\odot$) beyond ≈ 19 kpc [6]; independent stellar-halo number-density and σ_r profiles show a break radius $r_b \approx 16\text{--}27$ kpc [17, 18]. Together these give $\Delta r \lesssim 7.5$ kpc, i.e. $\Delta r/r_Z \lesssim 0.4$. Because of the

transition-radius model dependence (MOG $R_0 = 18.1$ kpc vs Newtonian $R_0 = 33.8$ kpc on the same data, N5) and FIRE-2 Jeans systematics of 5–20% (N6), this is an upper bound on the rotation-curve roll-off width, not a measurement of the field-theoretic width (gate T23a).

A15-T1b-1 — small-ratio candidates (RETRACTED-in-session).

The v1.2 candidates $\Delta r/r_Z \in \{A = 0.080, 1/Q = 0.091, Z/Q = 0.182\}$ predicted $\Delta r \approx 1.5\text{--}3.3$ kpc. The proxy $\Delta r/r_Z \lesssim 0.4$ exceeds the largest by $\approx 2.3\times$ and the smallest by $\approx 5.1\times$ (gate T23b); we withdraw the simple-ratio hypothesis rather than carry one the data already disfavour.

A15-T1b-2 — smooth-edge no-go (DERIVED-CONDITIONAL / toy ansatz).

A phenomenological smooth edge $\rho_\theta \propto r^{-2} e^{-r/r_c}$ normalizes cleanly: $M_{\text{total}} = 2\sigma^2 r_c / G$ forces $r_c = G M_{\text{total}} / (2\sigma^2) = r_Z$ exactly (gate T24a), a result worth registering. But this ansatz is not a solution of $\square\theta = 0$: with $\partial_r \theta \propto e^{-r/2r_c}/r$, the cylindrical Laplacian gives $\square\theta = -e^{-r/2r_c}/(2r r_c) \neq 0$ (gate T24b), implying a hidden source or damping. Moreover its slope transition is broad: γ runs from -0.1 to -0.4 over $\Delta r/r_Z \approx 2.2$ (gate T24c), about $5\times$ wider than the observed proxy of ≈ 0.4 (gate T24d). Hence $\Delta r/r_Z \approx 0.4$ is NOT recovered by any smooth massless-edge ansatz; the earlier v1.3 statement that a roll-off of order r_Z “matches” the data is corrected here — the smooth edge is in fact too gradual, and the data are sharper.

A15-T1b-3 — Sharp-Edge Mechanism (Path A upgraded, DERIVED-CONDITIONAL).

The open problem is not “which small ratio gives 0.4” but: why is the physical Region-II $\rightarrow$ Region-III matching sharper than a smooth massless edge? A scale census settles the direction. The only locked lengths in the problem are the Region-I radial-mode scale $\xi \approx 31 \ell_P$ ($\xi/r_Z \approx 9 \times 10^{-55}$), the boundary location r_Z itself ($r_Z/r_Z = 1$), and the infrared log scale $L \sim$ Hubble radius ($L/r_Z \approx 4 \times 10^5$). None equals ≈ 0.4 : a genuine field scale is either $\ll r_Z$ (sharp) or $\gg r_Z$ (gradual), so the observed $\Delta r/r_Z \lesssim 0.4$ selects the SHORT-scale (sharp) branch (gate T25a).

Path A (FRW point matching) — upgraded. In the corpus the Region-II/III matching is fixed by continuity of enclosed mass, $M_g(r_Z) = M_{\text{total}}$ (ZS-A1 §2.4.2): it is a global point boundary condition, not a local damping length r_c . Imposed on the scale-free $1/r^2$ interior, a point condition produces a near-step edge, $\Delta r/r_Z \ll 1$ (gate T25b), governed by the short scale rather than the IR scale. This introduces no new parameter and resolves the v1.4 “tension” constructively: the fact that the data are $\approx 5\times$ sharper than a smooth massless edge is not an anomaly but the PREDICTION of short-scale point matching (gate T25c). The qualitative claim — the intrinsic edge is near-step, not a smooth $O(r_Z)$ roll-off — is therefore DERIVED-CONDITIONAL (conditional on the matching being short-scale/topological, as the mass-exhaustion boundary condition implies). The corresponding paths B (smooth finite-line crossover with a simple geometric n) is DISFAVOURED — it is the wrong, too-gradual picture — and path C (observational upper bound) MERGES with A: the observed $\Delta r/r_Z \lesssim 0.4$ is the resolution-limited upper bound on an intrinsically sharp edge.

What stays open and what is now falsifiable. No numerical value of $\Delta r/r_Z$ is claimed; by the No-Intrinsic-Width Lemma the bulk equation fixes none, and the short-scale census only bounds it small. The new content is a falsifier (F-A15.7): if Gaia DR4 robustly RESOLVES a wide intrinsic transition, $\Delta r/r_Z \sim O(1)$, the short-scale point-matching prediction of Path A is falsified — turning the previously open sharpness question into a registered, pre-DR4 observational gate.

§6. Zero-Free-Parameter and Anti-Numerology Audit

Structural claim. Equation (3), the Goldstone origin of $\rho_\theta \propto 1/r^2$, and Region III use only A, Q, (Z, X, Y); nothing is tuned (gates T3–T6, T10, T13). Numerical value. Equation (2) is an identity, not a fit, so no candidate-scan Monte Carlo applies; the near-tautology of §5 is why the registered discriminant (§4) is the out-of-sample saturation-versus-growth behaviour and why A15-T1b is OPEN: the No-Intrinsic-Width Lemma (§5) shows $\square\theta=0$ fixes no width coefficient, so no candidate-scan applies. r_Z is labelled DERIVED-CONDITIONAL throughout. Cosmological consistency. $\Delta N_{\text{eff}}(0) = 0$ exactly in FRW (ZS-F1 §7.1), so the Planck 2018 background is unperturbed (T14); the only internal correction is the $\sigma \leftrightarrow v_{\text{flat}} (\sqrt{2})$ fix in ZS-A11 §3.2, affecting L only (T6).

§7. Scope and Non-Claims

NC1. A finite outer cutoff is DERIVED upstream (ZS-F1 §5.3) and shared by all finite-mass halo models; not novel (I1).

NC2. r_Z is conditional on observed M_{total} and v_{flat} ; no zero-parameter value prediction is asserted. The mathematical BVP is well-posed (A15-T1a); by the No-Intrinsic-Width Lemma the massless equation fixes no width coefficient, the observed $\Delta r/r_Z \lesssim 0.4$ is an upper bound (A15-T1b-0), and Path A (point matching) makes the intrinsic edge near-step (A15-T1b-3, DERIVED-CONDITIONAL).

NC3. A15 does not claim that all particle dark-matter models are excluded by a declining MW rotation curve. The frozen gate targets the standard extended-growth (M_{200} -scale, NFW-like) interpretation. Truncated or cored particle halos remain comparator models, penalized only by their additional truncation/profile freedom (N1, N3). On the rotation curve alone the protocol does not separate ε -Halo from declining MOG (N2).

NC4. By F-A5.7, a DR4 outcome that falsifies the ε -Halo outer boundary does not by itself confirm particle dark matter; it removes ε -Halo at the outer boundary and leaves the remaining classes to be separated by the independent channels of §8.

§8. Conclusion and Future Work

We have frozen and hash-locked, ahead of Gaia DR4 (2 December 2026), a single decisive Milky Way prediction: the enclosed mass saturates near $2 \times 10^{11} M_\odot$ by ≈ 25 kpc, so $\gamma \rightarrow -1/2$ and is maintained outward, versus a particle halo growing toward $M_{200} \approx 8 \times 10^{11} M_\odot$ to ≈ 190 kpc. $\gamma > -0.3$ beyond 25 kpc, or a still-growing enclosed mass, falsifies the ε -Halo outer boundary. The test establishes finite versus extended-growth particle-NFW; it cannot alone separate ε -Halo from tuned cored halos or declining MOG (§2A, §4B).

Two extensions are flagged. Closing A15-T1b (§5) would lift the value prediction above its conditional status. A companion paper, ZS-A16 (degeneracy-breaking), is reserved to add the independent channels that the no-go ledger shows are required: external-galaxy outer declines, stellar streams and satellite kinematics, the vertical force K_z , weak-lensing mass normalization, the CGM baryon-mass constraint, and an explicit profile-complexity penalty against cored-NFW and MOG. A15 is the DR4 pre-registration; A16 is where the cored-NFW and MOG degeneracies left OPEN here are broken.

Acknowledgements, Code and Data Availability

This paper consolidates internal Z-Spin Collaboration research notes. The frozen protocol (a15_dr4_protocol.yaml, SHA-256 1d04ee2f73b6f7791586fa3964d201e2336e54f690a75fc25f3ee0f773b283e2) and the 36-test verification suite (verify_zsa15.py) are available at <https://github.com/KennyKang-git/zspin>. All observational inputs are from the cited public literature and the forthcoming Gaia DR4 archive.

Appendix A. Worked SIS Lemmas

Isothermal hydrostatic equilibrium with $P = \rho\sigma^2$ and $dM/dr = 4\pi r^2\rho$ gives $\rho = \sigma^2/(2\pi G r^2)$, $M(<r) = 2\sigma^2 r/G$, $v_c^2 = 2\sigma^2$ (flat). Beyond saturation the point-mass limit gives $v_c \propto r^{-1/2}$ ($\gamma = -1/2$). Equating $M_P^2/(2L^2r^2)$ to $\sigma^2/(2\pi G r^2)$ gives $L^2 = \pi G M_P^2/\sigma^2$; with $\sigma \approx 156$ km/s, $L \approx 3.3 \times 10^{26}$ m.

Appendix B. Observational Input Compilation

Table B.1. Milky Way measurements relevant to the DR4 gates.

Study	Key quantity	Bearing on the gates
Jiao et al. 2023	$M_{\text{total}}=2.06 \times 10^{11}$; onset ≈ 19 kpc; flat rejected 3σ	Supports low-mass reading; $r_Z \approx 18$ kpc
Ou et al. 2024	faster decline to ≈ 30 kpc; cored Einasto	Decline confirmed, DM retained (N1)
Zhou et al. 2022	$M_{200}=8.05 \times 10^{11}$; $R_{200} \approx 192$ kpc	Full-halo reading; would falsify G-DR4.1
Cepheid Gaia DR3 2025	$V_c(R_\odot)=236.8 \pm 0.8$ km/s	Shifts r_Z band toward ≈ 16 kpc
Klacka & Sturc 2025	decline may be Jeans artifact	Motivates G-DR4.3 (N4)
LZ 2025 (417 d)	no WIMP 3–9 GeV; v-floor reached	F-A5.7 currently PASS (asymmetric)

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Version History

v1.5 (March 2026): Upgraded Path A of the Sharp-Edge Mechanism (A15-T1b-3) to DERIVED-CONDITIONAL with no new parameter. A scale census shows none of the locked lengths ($\xi/r_Z \approx 9 \times 10^{-55}$, $r_Z/r_Z = 1$, $L/r_Z \approx 4 \times 10^5$) equals the observed ≈ 0.4 , so the edge must be short-scale; the corpus matching $M_\varepsilon(r_Z) = M_{\text{total}}$ is a global point boundary condition that, on the scale-free interior, yields a near-step edge ($\Delta r/r_Z \ll 1$). This recasts the v1.4 “data sharper than the smooth field” tension as a prediction, disfavors path B, merges path C, and registers a new falsifier F-A15.7 (a wide DR4-resolved intrinsic width refutes it). The numerical $\Delta r/r_Z$ stays unclaimed (No-Intrinsic-Width Lemma). Verification suite 33 $\rightarrow$ 36 tests (T25a–c); SHA-256 re-locked. v1.4 (March 2026): Proved the No-Intrinsic-Width Lemma — $\square\theta = 0$ on a source-free annulus fixes $\theta = a \ln r + b$ and determines no universal $\Delta r/r_Z$ coefficient (PROVEN, gate T24). Corrected the v1.3 “roll-off matches the data” reading: the smooth edge $\rho \propto r^{-2}e^{-r/r_c}$ gives $r_c = r_Z$ exactly but is non-harmonic and $\approx 5\times$ too gradual. Restructured §5 into A15-T1b-0/1/2/3. Verification suite 29 $\rightarrow$ 33 tests (T24a–d). v1.3 (March 2026): Split A15-T1b into A15-T1b-0 (Gaia DR3 width proxy) and A15-T1b-1 (geometric quantization); RETRACTED {A, 1/Q, Z/Q}. Added N5, N6; suite 26 $\rightarrow$ 29 (T23a–c). v1.2 (March 2026): Added §2A Imported Theorem and No-Go Ledger (I1–I3, N1–N4); split A15-T1 into A15-T1a (IMPORTED-PROVEN) and A15-T1b; §4B five-truth confusion-matrix; suite 22 $\rightarrow$ 26. v1.1 (March 2026): renumbered ZS-A13 $\rightarrow$ ZS-A15; added §4A frozen hash-locked pipeline, §4B decision matrix, §5 Open Gate A15-T1; softened NFW wording (NC3); suite 16 $\rightarrow$ 22. v1.0 content preserved; changes are additive.